

Securing the Edge: Analyzing and Mitigating False Data Injection in Industrial IoT (IIoT) Gateways

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CS 671: Cybersecurity

February 19, 2026

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Problem Statement

This project addresses the critical threat of False Data Injection (FDI) in Industrial IoT (IIoT) and Operational Technology (OT) systems. As legacy industrial networks connect to the internet via IIoT gateways, protocols like Modbus/TCP—which often lack encryption—become vulnerable. We will analyze how an attacker can manipulate sensor data at the gateway level to cause physical system failures or incorrect automated responses in critical infrastructure.

Motivation & Related Work

As a student at California State University, East Bay, I am aware that one of the top concerns for modern computer engineers is safeguarding the hardware-software interface. Prominent incidents like the Oldsmar Water Plant hack demonstrate that changing even a few data points can have physically lethal consequences. In the context of related work, prior research into legacy vulnerabilities highlights that many Operational Technology (OT) systems are insecure by design because they historically prioritize operational uptime over robust encryption. While standards such as NIST SP 800-82 provide essential guidelines for industrial security, real-world implementations often struggle to reconcile these requirements with the significant resource constraints of IIoT hardware.

High-Level Description of Idea

We will build a virtualized testbed to demonstrate a detailed analysis of an IIoT threat.

Technical Approach

Our approach involves a multi-staged simulation environment centered on Mininet-IIoT to emulate a robust network consisting of an IIoT sensor, a PLC gateway, and an integrated attacker node. We will execute the threat analysis by performing a Man-in-the-Middle (MitM) attack using Python and the Scapy library to intercept and modify Modbus register values in real-time, effectively demonstrating how sensor data can be manipulated. To counter this, we will develop and implement a physics-aware Intrusion Detection System (IDS) in Python that serves as our primary defense, designed to trigger alerts whenever data values shift at a rate that is physically impossible for the emulated hardware to achieve. The technical execution will rely on a specific toolkit including Python for programming via Scapy and Pymodbus, Wireshark for detailed packet capture and analysis, and an Ubuntu Linux environment to host the Mininet-IIoT simulation.

Evaluation Metrics

To accurately assess the effectiveness of the proposed physics-aware Intrusion Detection System (IDS) and the impact of the False Data Injection (FDI) attack, we will utilize the following evaluation metrics:

- **Detection Accuracy:** The IDS will be evaluated using standard classification metrics, primarily focusing on True Positive Rate (TPR), False Positive Rate (FPR), Precision, Recall, and the F1-Score. Low false positive rates are critical to avoid unnecessary and costly system shutdowns.
- **Detection Latency:** This measures the time elapsed between the injection of manipulated Modbus data by the attacker and the generation of an alert by the IDS. In Operational Technology (OT) environments, this latency must be kept to milliseconds to prevent automated systems from reacting to the malicious data and causing physical damage.
- **System Overhead:** We will measure the CPU and memory utilization of the Python-based IDS. Because real-world IIoT edge gateways are often resource-constrained devices (e.g., Raspberry Pis or specialized industrial routers), the defensive solution must demonstrate a lightweight footprint.

Simulation Realism vs. Real-World IIoT Systems

While the Mininet-IIoT testbed provides a highly controlled and safe environment for executing destructive threat analyses, it balances high-fidelity emulation with the inherent limitations of virtualization.

- **Protocol and Network Fidelity (High Realism):** The simulation achieves a high degree of realism at the network layer. By utilizing libraries like Pymodbus and Scapy, the generated Modbus/TCP traffic contains authentic packet structures and payloads. Mininet-IIoT utilizes real Linux network namespaces, meaning the virtualized IP stacks, routing behaviors, and bandwidth constraints closely mirror those of a physical IIoT network. The network traffic captured will be indistinguishable from a physical network when analyzed in Wireshark.
- **Hardware and Firmware Abstraction (Simulation Limitation):** The primary divergence from reality lies in the hardware layer. Our simulated PLCs and sensors run as software processes. Consequently, the testbed will not replicate hardware-specific timing issues, proprietary firmware vulnerabilities, or physical clock drift that an attacker might exploit in physical hardware.
- **Physical Process Dynamics (Simulation Limitation):** The industrial environment (e.g., a water tank or turbine) will be simulated mathematically via Python scripts. While our physics-aware IDS will successfully demonstrate the concept of checking data against physical bounds (e.g., water cannot boil in one second), it relies on a simplified deterministic model. Real-world physical environments introduce chaotic background noise, mechanical degradation, and complex thermodynamics that are difficult to perfectly replicate in a lightweight containerized simulation. Overall, the simulation provides high realism at the network and protocol level but is limited in accurately modeling hardware behavior and complex real-world physical processes.

Expected Challenges

One of the primary technical challenges for this project is the virtualization overhead associated with running a full network simulation simultaneously with a heavy load of four other classes. To address this resource constraint, the team will utilize lightweight Docker-based containers wherever possible to minimize CPU and RAM consumption. Another critical research challenge involves maintaining technical accuracy to ensure that the simulated Modbus traffic accurately reflects the behavior of real-world industrial hardware. We plan to mitigate this by strictly validating our generated traffic against industry-standard PCAP files and protocol specifications to ensure the simulation remains a high-fidelity model for cybersecurity analysis.

Tentative Timeline

The following table outlines the schedule for the project:

Table 1

Project Phases and Milestones

Week	Phase	Milestone
Week 6	Setup	Finalize Mininet-IoT topology for "Security Solutions."
Week 7	Baseline	Capture normal Modbus/TCP traffic.
Week 8	Attack	Successfully inject false sensor data into the stream.
Week 9	Defense	Develop and test the Python-based IDS.
Week 10	Report	Finalize the comprehensive analysis for the professor.